HATCHSIDE · PRECIS

HatchSide design precis

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-03	Amish Chadha	TRL 3; constructable design (HTS-DDR-002), numbers from HTS-CAL-001, build plan and decisions register
0.2	2026-10-03	Amish Chadha	TRL 2; how it works, components, design choices, first-order numbers, concept media (HTS-DDR-001)
0.1	2026-09-30	Amish Chadha	Initial scaffold

HatchSide design precis

Lets crews clean, sample and retrieve through a manhole or tank hatch from a tripod, so nobody enters; includes the DomeReach tool head for household biogas digesters.

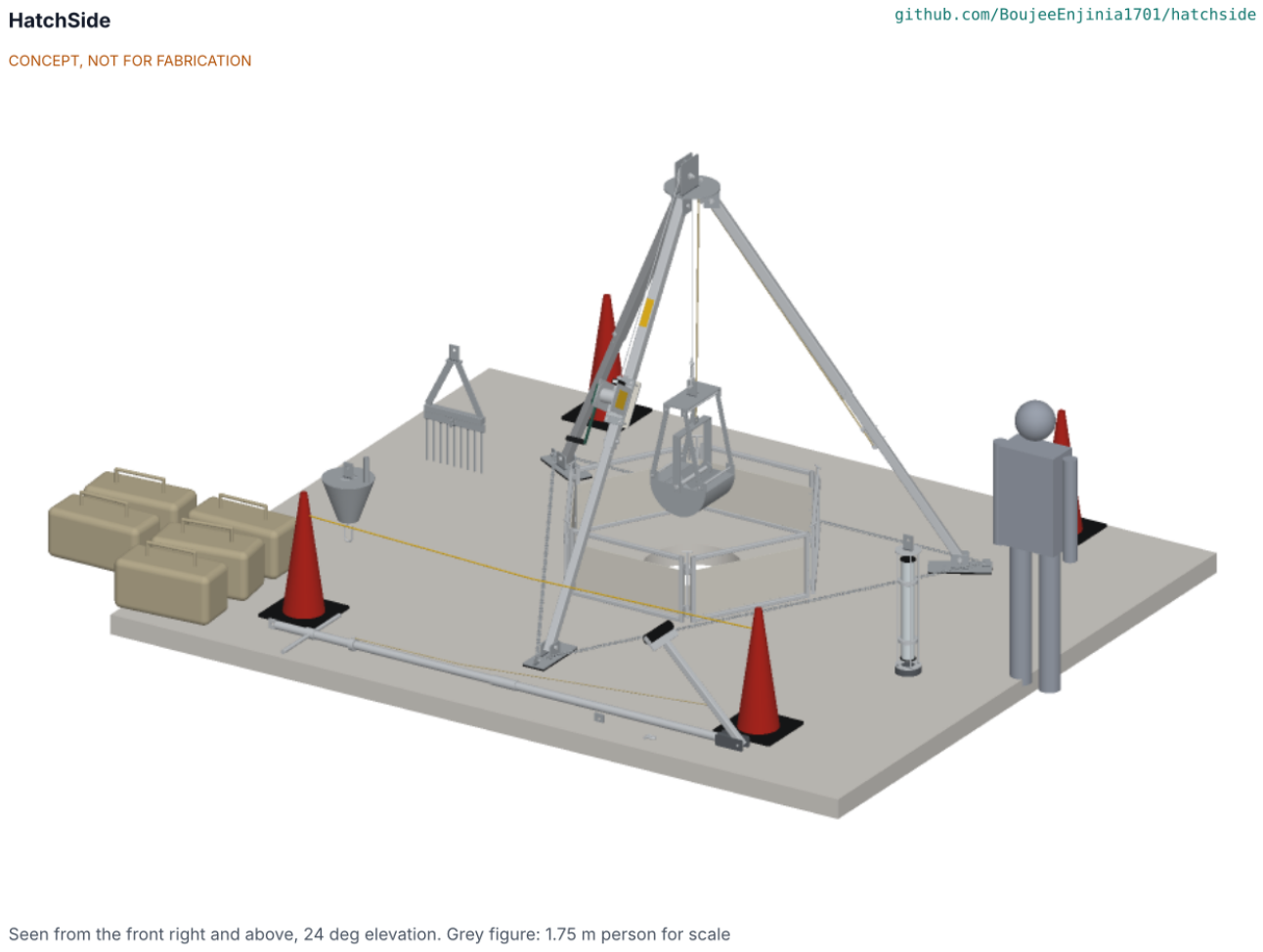


Figure 1. HatchSide set up over a 560 mm street manhole inside its rim guard, with the grab on the rope and the other four heads on the ground; grey figure 1.75 m for scale. Concept, not for fabrication.

How it works

The crew cones off the site, lifts the cover and stands a three-legged aluminium tripod over the opening. The legs are already bolted to a steel head plate; each telescopes and locks with a ball-lock pin, and three chains between the feet stop them spreading. A two-speed hand winch with an automatic load brake is clamped to leg A. Its 6 mm stainless rope runs up beside leg A, over a 150 mm sheave held between two steel cheek plates on the head, and drops straight down the centre of the opening. On the rope's end a swivel and a screw-pin shackle take any of five tool heads, all of which carry the same 10 mm steel tab with a 20 mm hole.

- **Grab:** a clamshell that is lowered open and closed by a working line led through a snatch block under the head and tied off on a cleat on leg B; it lifts about 10.8 L of wet silt per cycle (estimate).
- **Sampler:** a clear tube in a ballasted cage, with caps tripped shut at a chosen depth read from a line counter.

- **Rake:** a ballasted tine bar dragged through scum or a blockage by the working line.
- **Retrieval head:** a funnel steered by a pole so its auto-locking connector clips onto a line already on a casualty; the haul is made with certified rescue equipment.
- **DomeReach:** a two-section pole with a single hinged arm worked by a pull line, for breaking scum, lifting sludge and rolling generic sealant inside a household biogas digester from its opening.

Six hinged panels pinned into a hexagon round the opening keep people and traffic out and splash in. Every job is done with the crew standing outside the guard.



Figure 2. Cutaway with the front half removed: the grab lowered 1.6 m down the manhole shaft. Concept, not for fabrication.

Components

Table 1. Main components (numbers match `bom/bom.csv` and Figure 3).

BOM	Component	Role
1	Feet with clevis, chain lug and rubber pad (3)	Bear on the street and anchor the chains
2, 3	Lower and upper legs, square aluminium tube (3 each)	Telescoping legs, 2,585 mm pin to pin at normal length
4	Tripod head with mast cheeks	Hinges the legs and carries the sheave
5	Head sheave and M20 axle	Turns the rope down the centre line
6, 7	Hinge and foot bolts, adjust pins	Join the legs to the head and feet; set leg length
8	Leg-spread chains (3)	Stop the feet spreading
9, 10	Winch bracket and hand winch	Raise and lower heads; the brake holds the load
11, 12	Wire rope and line counter	15 m of rope; depth read to 0.1 m
13	Head interface (swivel, shackle, gauge plate)	The one published coupling for every head
14	Snatch block, working line and cleat	Close the grab, drag the rake, trip the sampler
15, 16	Rim guard panels and drop pins	Barrier and splash guard round the opening
17 to 21	Grab, sampler, rake, retrieval and DomeReach heads	The five tasks
22 to 24	Cones and chain, carry bags, labels	Work zone, transport, markings

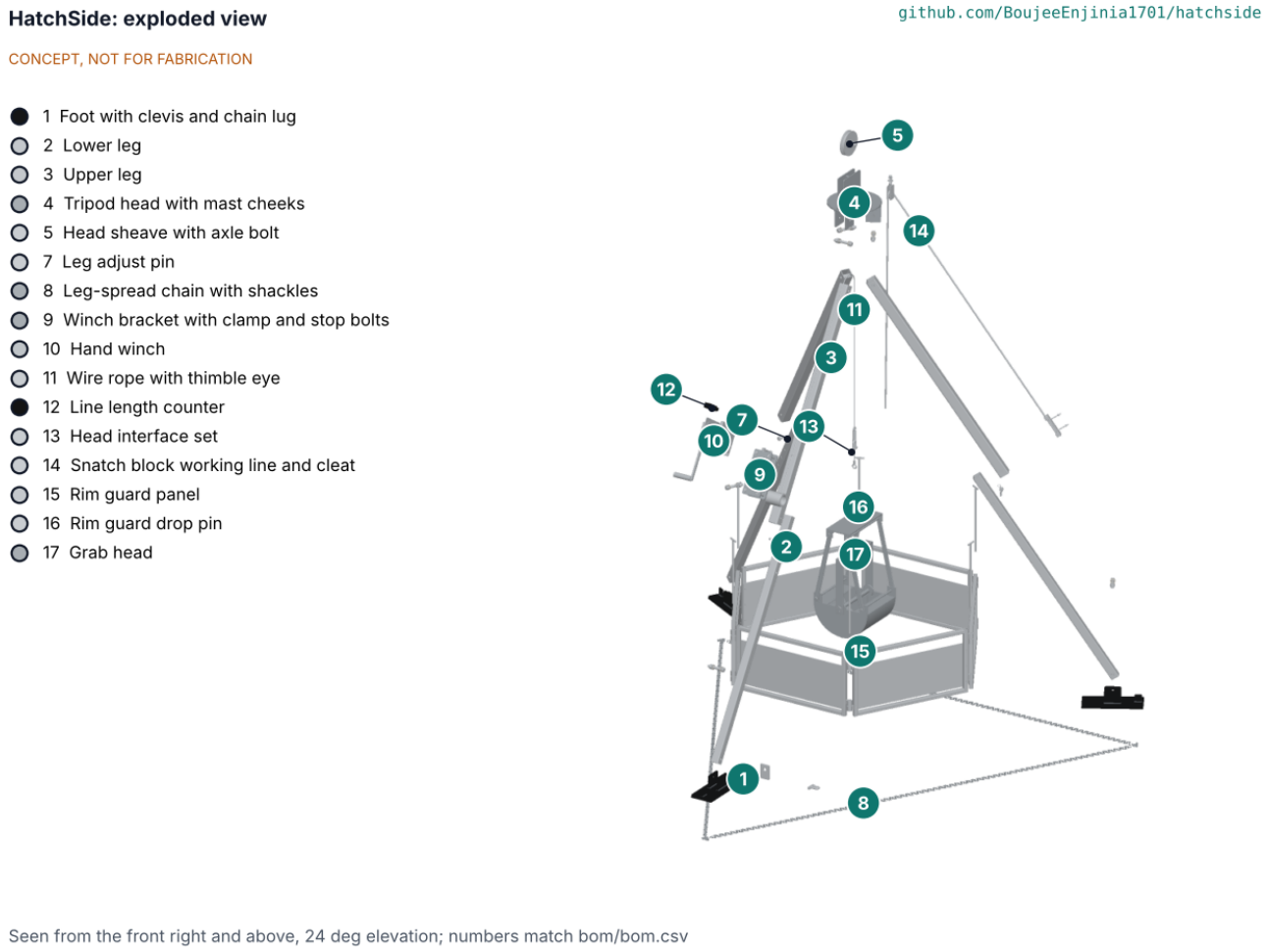


Figure 3. Exploded view with BOM numbers. Concept, not for fabrication.

Key design choices

All decided by Amish under his 2026-10-03 pre-approval (HTS-DDR-001 and HTS-DDR-002; index in HTS-DEC-001).

- **Material handling only, 150 kg.** The frame is rated for tool heads, not people. This keeps every pack under 25 kg and keeps the design out of personnel-rescue certification.
- **No separate mast.** The sheave sits between two cheek plates welded to the head plate, at the height where the rope from the winch runs parallel to leg A. One welded part replaces a mast, its joint and its bracing.
- **Positive winch load path.** The winch bracket clamps round leg A but bears on a through stop bolt, so the rope pull never relies on friction on aluminium.
- **One head interface.** A 10 mm tab with a 20 mm hole 30 mm below its top edge. Anyone can make a new head that fits.
- **Hand power and single hinges.** No engine, no powered arm and no gripper; the grab and DomeReach are worked by lines from the rim (patent design-around for articulated grippers).
- **Galvanized steel, 6082 aluminium and A4 stainless,** with nylon isolation between steel and aluminium.

First-order numbers

From HTS-CAL-001 (estimates; nothing measured).

Table 2. First-order numbers.

Quantity	Value	Assumption
Sheave height above ground	2.41 m	Legs at 32.5 degrees, feet on a 3.0 m circle
Leg force, winch leg, at 150 kg x 1.5 x 1.25	3.85 kN	Rope runs along leg A
Leg buckling margin	5.5	Whole leg as the 50 x 3 mm section
Rope safety factor at 150 kg	14.3	21 kN minimum breaking load
Tipping angle of the rope at 150 kg	25 degrees	74.7 kg tripod, 750 mm to the tipping edge
Crank force at 150 kg, low gear	31 N	10:1, 250 mm crank, 75 % efficiency
Grab payload per cycle	10.8 L (19 kg)	70 % fill, 1.8 kg/L
Silt removed per hour at 3 m	About 166 L	3.9 minutes a cycle
Heaviest packed load	23.7 kg	Grab head
Setup time, two people	9.0 minutes	Legs travel bolted to the head
Estimated parts cost	USD 2,838	`bom/bom.csv`

Value-engineering target: USD 5,000. Estimated cost of the constructable design: USD 2,838 (USD 2,162 under the target).

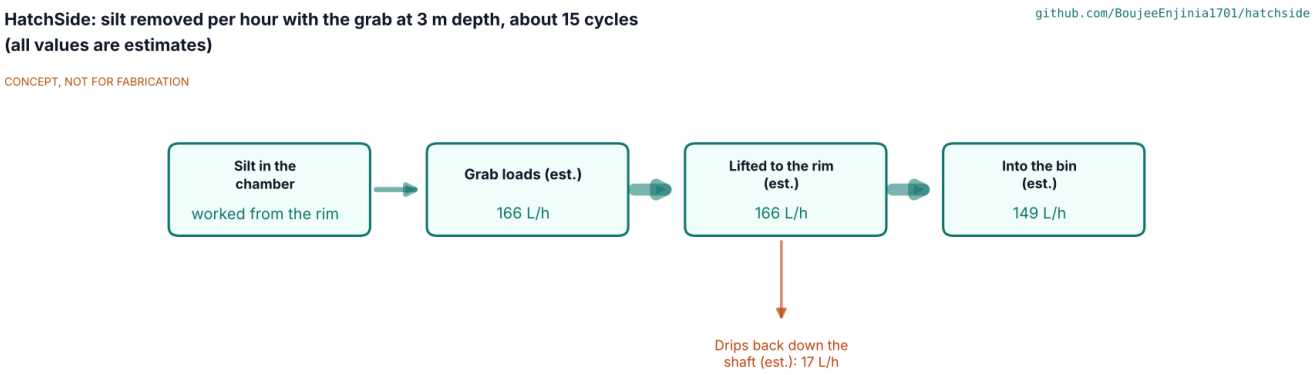


Figure 4. Silt removed per hour with the grab at 3 m depth (all values are estimates).

Patent design-arounds

From the preliminary patent, trademark and prior-art screen (not legal advice):

- No articulated arm with a multi-finger gripper (Genrobotics Bandicoot patents); DomeReach has one hinge and no gripper.

- The name TopSide was dropped because of a generic mark and the US Navy TOPSIDE C3 mark; HatchSide is used instead.
- DomeReach names its sealant generically, never by brand.
- If the tethered crawler block is later adopted, it uses a generic published head interface (connector plus bolts), no adjustable crawler legs and a simple payout encoder, per the 2026-09-30 screen.

Shared blocks

- Tripod and mast block: at 150 kg it is not shared with WellSling, whose 400 to 600 kg loads need their own frame (HTS-DDR-001, D1).
- CalRig proof-load (winch, rope and tripod proof testing) at TRL 4.
- Tethered crawler block with CulvertCrawl and VoidScope, if adopted later.

Safety

Safety: Safety-critical: HatchSide lifts loads over confined spaces. It is published as an open engineering reference, never as certified rescue, lifting or fall-protection equipment.

- Never enter the space. The kit exists so that nobody has to. The winch is a material winch: never use it to lift or lower a person.
- The retrieval head only clips a connector onto a line already on a casualty. The haul is made with certified rescue equipment by trained rescuers.
- Test the air at the opening with a certified gas detector before and during work; gas can escape at the rim.
- Keep sparks and flames away from sewers and biogas digesters; methane can ignite.
- Guard the open hatch from traffic and falls, keep the rope within 10 degrees of vertical, and never stand under a raised load.
- Proof-load the tripod, winch and rope to 1.5 times the safe working load with a competent person before first use over an opening.

This design is not certified equipment.

Where to go next

- Prototype build plan: 05-build-plan.md (HTS-BLD-001).
- Design decisions register: 06-design-decisions.md (HTS-DEC-001).
- Calculations: 04-calcs/01-sizing.md (HTS-CAL-001).